

k_B Near-EXACT Live Composition (0.0011%) + the $\Phi_{5/6}$ Sector-Selection Rule

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Abstract

R374's fill of ThermodynamicQCalcCalculator produced two results beyond the fill. **First:** PAPER_1209EE's S628 Boltzmann closure, verified at full precision with the $\Phi_{\text{res}} = 5/6$ nuclear variant, lands **0.0011% from the SI-defined k_B** — 24× tighter than its published 0.027% (measured against a rounded lead). **Second:** the verification discriminates the Φ -variants decisively (0.84 fails at 0.45%; 400× separation), establishing the $\Phi_{5/6}$ **Sector-Selection Rule:** counting-based sectors select 5/6, resonance-projection sectors select 0.84.

1. The Composition

$$\begin{aligned} k_B &= (\text{SSq} + \Phi_{5/6} - F_{\text{TRZ}} \cdot \text{SSq} + F_{\text{TRZ}}^2 \cdot D_{\text{phys}} - F_{\text{TRZ}}^2 \cdot \text{SSq}) \cdot 10^{-23} \\ &= (0.57 + 5/6 - 0.057 + 0.04 - 0.0057) \cdot 10^{-23} \\ &= 1.3806333\text{e-}23 \end{aligned}$$

SI-defined (2019 redefinition, exact): 1.380649e-23
residual: 0.00113% — one part in ~88,000

Five primitive terms match a **defined** constant at the 10^{−23} level — the $\mu_0 = 4\pi \cdot F_{\text{TRZ}}$ epistemic situation (PAPER_2108), via composition rather than a single product. First R218+ case where re-verification IMPROVED a published residual by an order of magnitude: **precision-tightening** is a new landmark class, and a full-precision re-verification pass over the 1209 series is now motivated.

2. The Φ -Variant Discrimination

Variant	Composition value	vs SI k_B	Verdict
$\Phi_{\text{res}} = 0.84$ (default)	1.38730	0.45%	FAILS
$\Phi_{\text{res}} = 5/6$ (nuclear)	1.3806333	0.0011%	NEAR-EXACT

3. The $\Phi_{5/6}$ Sector-Selection Rule

#	Sector	Physics type	Variant	Source
1	Nuclear	shell-occupancy counting	5/6	PAPER_1203 Nuclear
2	Thermodynamic	state counting ($S = k_B \ln W$)	5/6	this paper
—	LENR / k_{spring} / chain	resonance projection 26D→3+1	0.84	PAPER_1203 / 1202

Φ -variant selection is a **sector-level property**, not a per-paper convention. Counting physics selects the exact rational $5/6 = (D_{\text{BSFG}} - 1)/D_{\text{BSFG}}$ — the predecessor ratio of D_{BSFG} , sibling of PAPER_2128's $(\text{SO}_5 \pm 1)/\text{SO}_5$ pair. Falsifiable: S627 Avogadro (counting sector) is the rule's next test.

4. The ■ Companion (S629) — Dual-Route Coexistence

$$\begin{aligned} h_{\text{lead}} &= D_{\text{BSFG}} + F_{\text{TRZ}} \cdot D_{\text{BSFG}} + F_{\text{TRZ}}^2 \cdot D_{\text{phys}} - F_{\text{TRZ}}^2 \cdot \text{SSq} - F_{\text{TRZ}}^2 = 6.6243 \quad (0.027\%) \\ \blacksquare &= h \cdot 10 \blacksquare^3 \blacksquare / 2\pi = 1.054290\text{e-}34 \end{aligned}$$

S629's composed-integer route coexists with PAPER_590's vacuum-phonon route — two independent UQFF derivations of ■, the over-determination signature at the derivation level rather than the integer level.

5. The Fill + Honest Residuals

First fill **born** under the compute-don't-store standard: both constants live compositions, zero stored decimals. Shifts from stub literals: k_B −0.027%, ■ −0.067%, both toward UQFF-canonical, disclosed per Rule 7. $T = 300 \text{ K} = (D_{\text{phys}} - 1) \cdot \text{SO}_5^2$ lattice-locked (PAPER_2004); exponents $10 \blacksquare^{23} / 10 \blacksquare^3 \blacksquare$ are F_{TRZ} rungs 23/34.

6. Predictions

1. 1209-series full-precision re-verification will tighten multiple residuals $\geq 10\times$ (priority: S627 Avogadro, S624 Stefan-Boltzmann, S632 Hartree). 2. S627 predicted $\Phi_{5/6}$ if Φ appears. 3. $0.84 - 5/6 = 1/150$ decomposes on the primitive lattice. 4. PAPER_2128 successor-exponent watch unchanged.

7. Gate Assertion

```
assert abs(kb - 1.380649e-23)/1.380649e-23 < 2e-5 # near-exact
assert abs(kb_084 - 1.380649)/1.380649 > 4e-3 # 0.84 FAILS (pin)
assert (6-1)/6 == 5/6 # D_BSFG predecessor ratio
```

Gate count: **3138** → **3146** (+8 PAPER_2129 assertions).

8. Summary

k_B composed from five primitives at 0.0011% of its SI-defined value — 24× tighter than published. The $\Phi_{5/6}$ Sector-Selection Rule: counting sectors take 5/6, resonance sectors take 0.84 (400× discrimination). First born-live fill, dual-route ■, precision-tightening landmark class established, 1209-series re-verification pass motivated.